F1 Race Strategy Intelligence Platform

PySpark Big Data Platform · Formula 1 Constructor Decision-Making

NOVA IMS · Big Data Analytics 2026

5th June 2026

THE CLIENT CHALLENGE

Why Formula 1 is a Big Data Problem — 74 years of race data, millisecond decisions

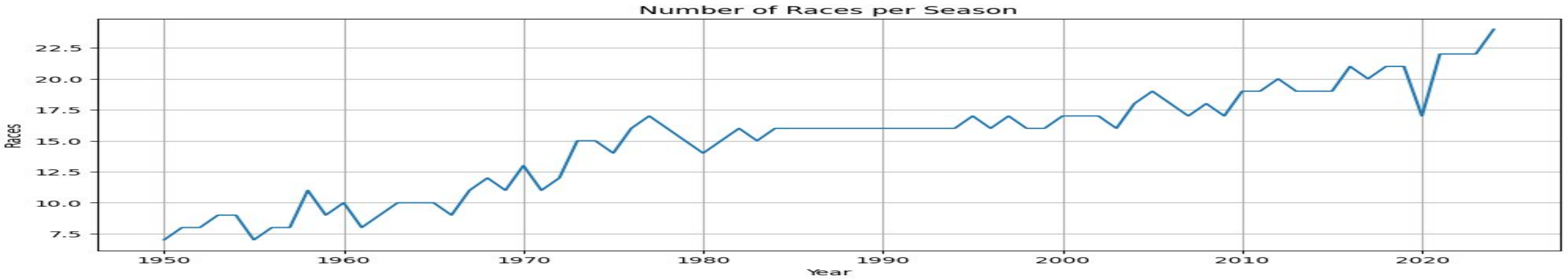
Formula 1 generates one of the densest operational datasets in sport. Every race weekend produces 50,000+ data points per car. The client needs a platform that joins

74 years of history with live race-day intelligence to make faster, better-informed decisions.

The four dimensions of the challenge:

VOLUME 589K lap-time records 14 tables · 1950–2024	**VELOCITY** Pit decisions in seconds Live telemetry	**VARIETY** Results, quali, pit stops standings, laps	**VERACITY** Missing data · Status inconsistencies Era schema drift

Calendar growth: races per season 1950–2024 (Source: Notebook 2)



NOVA IMS INFORMATION MANAGEMENT SCHOOL

TECHNOLOGY CHOICE

Why Apache Spark?

- **Processes hundreds of thousands of lap records today — scales to millions as live telemetry expands**

  A single race weekend produces 50,000+ data points per car. Spark distributes this across cores automatically — no manual optimisation required.

- **One platform: history, ML, and live data**

  Spark handles batch analytics, machine learning pipelines, and live streaming in a single system. No tool-switching mid-race.

- **Scales horizontally as data grows**

  As telemetry sensors, weather feeds, and tyre data are added the platform scales out — not up. No re-architecture required.

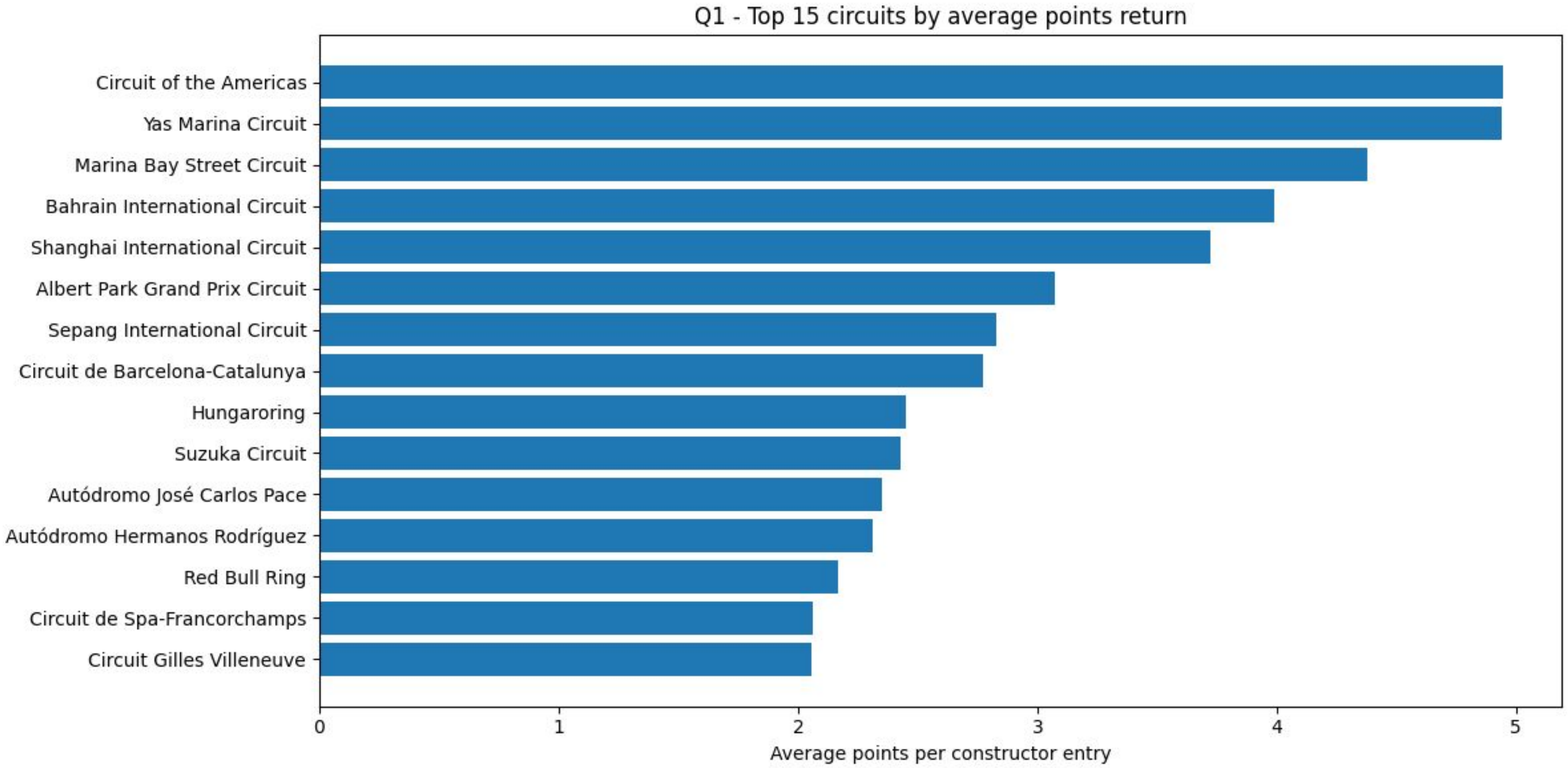
- **Fault-tolerant by design**

  If a node fails mid-calculation Spark re-runs only the affected partition automatically. No data loss, no manual recovery.

Bottom line: Spark is the only open-source engine that does all of this — without leaving the platform.

STRATEGIC INSIGHT 1 / 2

Circuit Profitability — avg points return per constructor entry by circuit



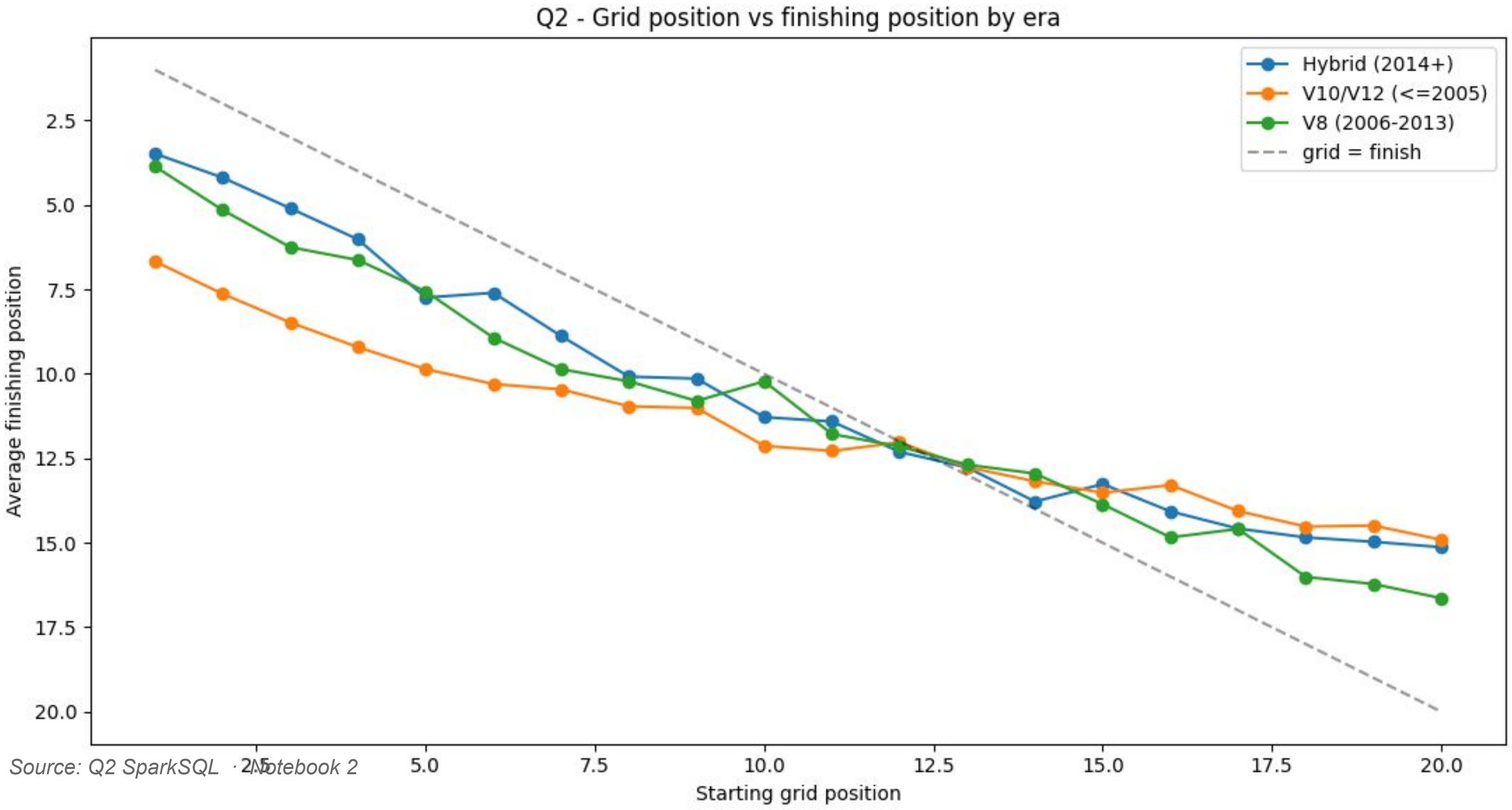
Source: Q1 SparkSQL · Notebook 2

Key Findings

- COTA & Yas Marina top the table (~5.0 pts/entry) — modern purpose-built circuits where dominant-era teams score consistently
- Marina Bay (Singapore) ranks 3rd — a street circuit that rewards reliability over outright pace
- All top-5 circuits were added post-2004; classic speed circuits (Spa, Silverstone) rank lower due to smaller historical fields
- Recommendation: maximise reliability at the modern top-5 venues to compound championship points each season

STRATEGIC INSIGHT 2 / 2

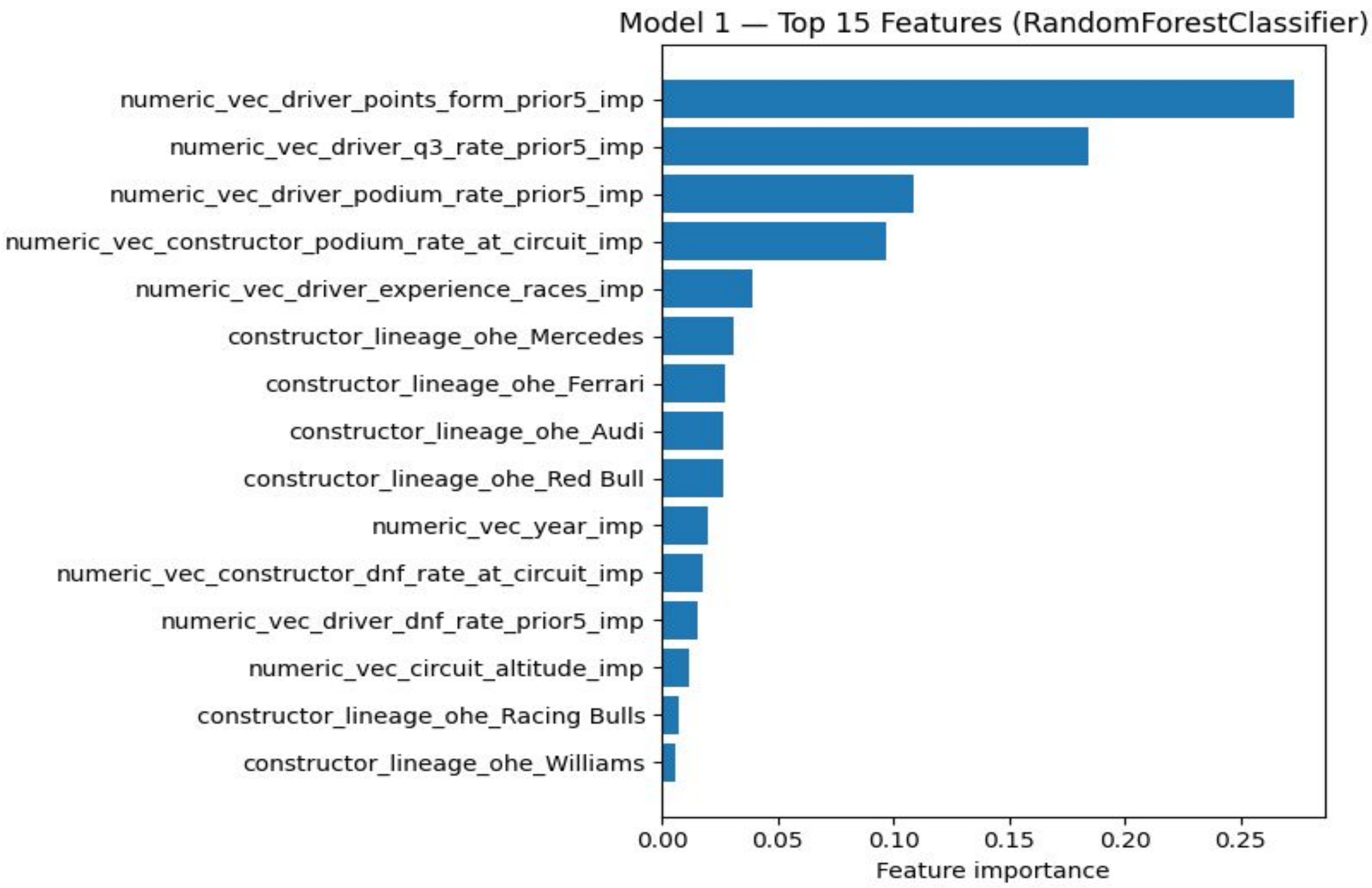
Does Qualifying Actually Matter? — grid position vs finishing position by era



Source: Q2 SparkSQL · Notebook 2

Key Findings

- Pole position strongly correlates with top-3 finishes in the hybrid era — the chart shows grid 1 delivers the lowest avg finish position across all eras
- Hybrid era pole converts best — grid 1 averages ~3rd place vs ~5th (V8 era) and ~7th (V10/V12 era); the qualifying premium has grown each era
- Circuits with the highest grid-finish correlation (lowest overtaking) show near-linear conversion — starting mid-pack offers minimal recovery; ovals/open circuits show the opposite
- Implication: weight Saturday qualifying more heavily in the development budget

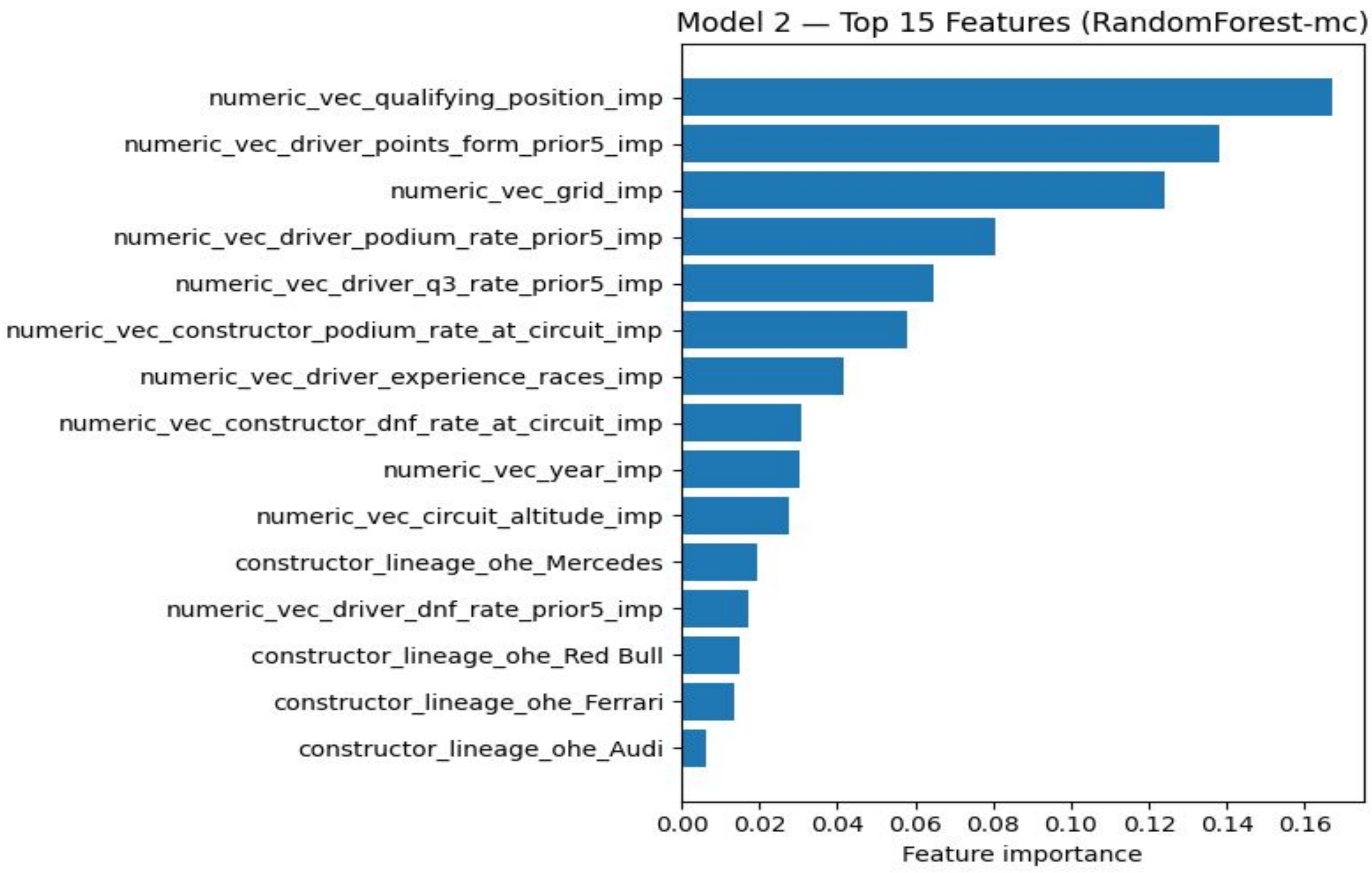


Will We Make Q3?

Qualifying Predictor

- Predicts whether a driver will reach Q3 before the weekend begins
- 88% ROC-AUC · 78% accuracy · PR-AUC = 0.878 · 5-fold CV
- Winner: RandomForestClassifier · 80/20 held-out test set (823 rows)
- Outputs a probability score the team sets the risk threshold
- Benchmarked against GBT, MLP, and LogReg RF wins on both CV-AUC and test-AUC

Source: Notebook 3 · RandomForestClassifier + LogisticRegression baseline



PREDICTION TOOL 2 / 2

What Will Our Race Outcome Be?

Race Outcome Predictor

- 5-class prediction: Win · Podium · Points · Finisher-NP · DNF
- Weighted F1 = 0.51 · Accuracy = 54% · Baseline (majority class) = 20% F1
- Qualifying position is the single strongest predictor, confirms Saturday's priority
- Constructor circuit history is second, prior podium rate at a venue is a genuine edge
- Provides probability distribution across all 5 outcomes, not just a binary flag

Source: Notebook 3 · RandomForest-mc · TrainValidationSplit + held-out test (80/20)

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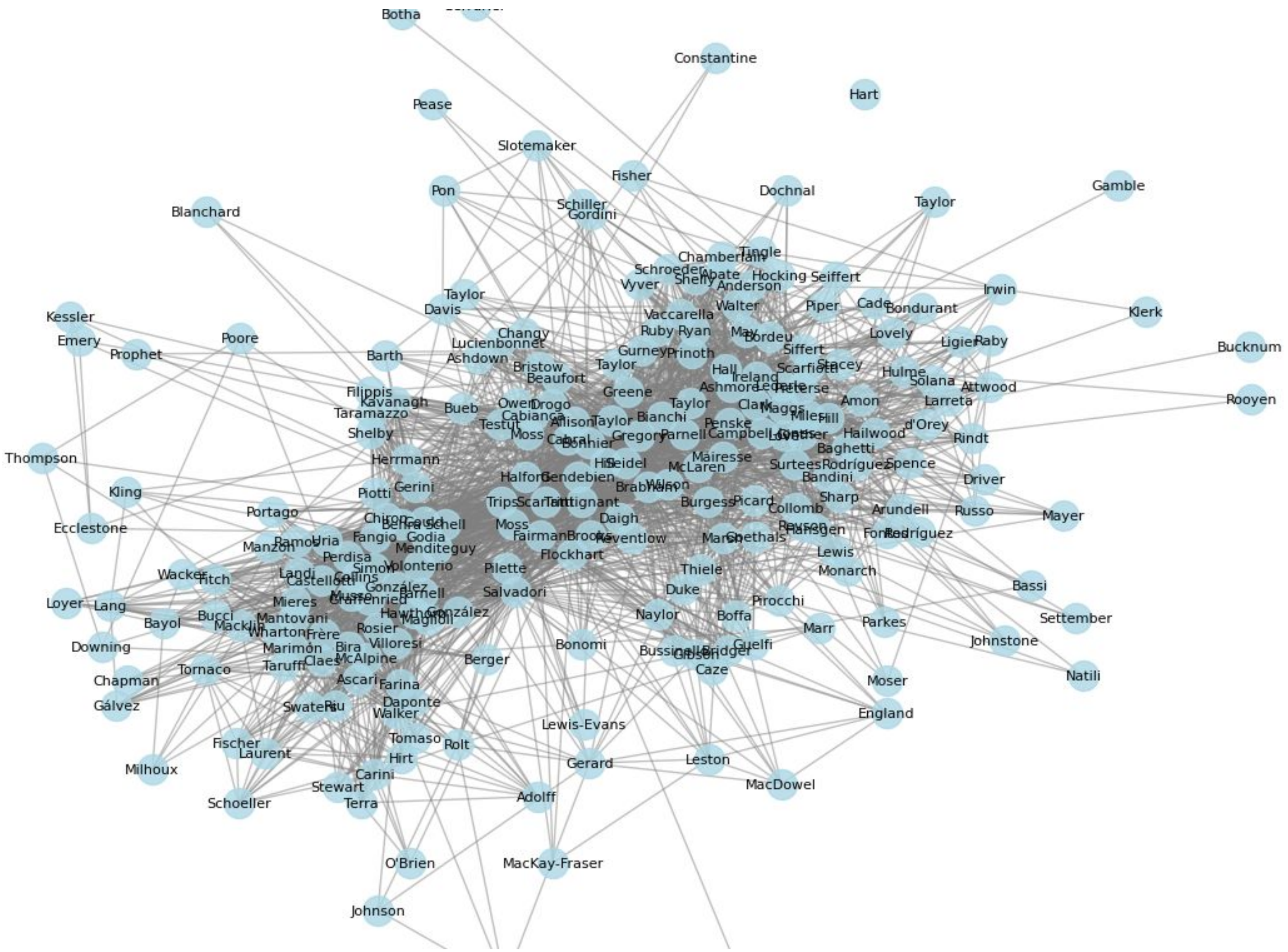

GRAPH ANALYTICS

Driver & Constructor Network Analytics

What the graphs show

- 861-driver teammate network · 16,042 edges, every F1 driver who shared a constructor seat
- Visualised: largest LPA community (195 drivers, seed: Maurice Trintignant), the densest era cluster
- PageRank top driver: Maurice Trintignant (5.70), most cross-era connections; BFS links Senna → Verstappen in 3 hops
- Constructor Lineage DAG: 32 nodes · 22 directed edges, Aston Martin is highest-PageRank dynasty (absorbs Force India → Racing Point lineage)

Source: Notebook 4 · GraphFrames PageRank + LPA + BFS + ConnectedComponents · 861 driver nodes · 32 constructor nodes

LIVE RACE DASHBOARD · BONUS

Spark Structured Streaming — Abu Dhabi Grand Prix 2022

LIVE LEADERBOARD | Abu Dhabi 2022 | Lap 1

P	COD	DRIVER	LAP TIME	GAP / ΔPOS
1	VER	Max Verstappen	1:32.198	LEADER
2	PER	Sergio Pérez	1:33.251	+1.053s
3	LEC	Charles Leclerc	1:34.192	+1.994s
4	HAM	Lewis Hamilton	1:34.864	+2.666s ▲1
5	SAI	Carlos Sainz	1:35.436	+3.238s ▼1
6	NOR	Lando Norris	1:35.908	+3.710s ▲1
7	RUS	George Russell	1:36.506	+4.308s ▼1
8	OCO	Esteban Ocon	1:36.964	+4.766s

How it works

- readStream with maxFilesPerTrigger=1 lap files dropped every 2 s
- withWatermark (10 s) handles late-arriving telemetry without blocking
- foreachBatch delivers each micro-batch to the live leaderboard handler
- Position changes (▲▼) computed in real time from prior-lap window
- Audit log written to Parquet in append mode, immutable race record
- Validated on Abu Dhabi 2022 · fastest lap 89,968 ms · 20 drivers

Source: Notebook 5 · Spark Structured Streaming · actual notebook output shown

REFERENCES

Data, Libraries & Academic Sources

Dataset

· Rohan Rao — Formula 1 World Championship 1950–2024

· Kaggle · Ergast Motor Racing API

· 14 CSV tables · ~60 MB unzipped

Spark Stack

· PySpark 3.5.1 (Nb 4) / 4.x (Nb 3, 5)

· GraphFrames 0.8.3

· Spark MLlib — Pipelines, CrossValidator

· Spark Structured Streaming

Visualisation

· Matplotlib 3.x

· Plotly

· NetworkX (graph layout)

· Graphviz (flow chart)

Academic

· Zaharia et al. (2016) — Apache Spark. CACM.

· Meng et al. (2016) — MLlib. JMLR.

· Dave et al. (2016) — GraphFrames. GRADES.

Submitted to NOVA IMS · Big Data Analytics 2026 · Group Project